Calculus Refresher

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Contents

Personal Notes

These notes are a concise collection of derived problems inspired by the examples from the Calculus books by Marsden and Weinstein.

Chapter names in these notes follows the chapter headings in the book.

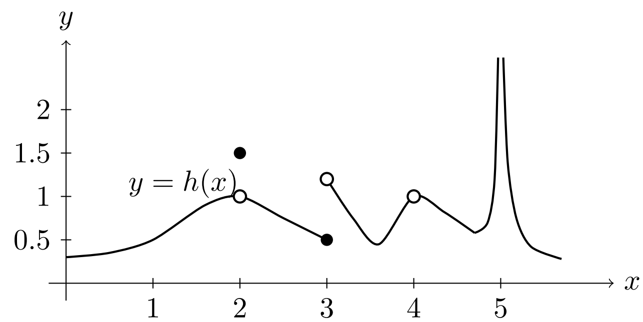
1 Derivatives and Limits

1.1 Reading Limits from a Graph

From the graph of $y = h(x)$ below, find $\lim_{x\to a} h(x)$, if it exists, for

$$a = 1, 2, 3, 4, 5.$$

Also state $h(a)$ whenever it is defined.



Answer

From the graph we read:

$$\lim_{x\to 1} h(x) = 0.5$$

since the graph is continuous there.

$$\lim_{x\to 2} h(x) = 1$$

even though

$$h(2) = 1.5.$$

At $x = 3$, the left-hand limit is 0.5 while the right-hand limit is 1.2, so

$$\lim_{x\to 3} h(x)$$

does not exist.
At $x = 4$, the graph approaches 1 from both sides, so

$$\lim_{x\to 4} h(x) = 1,$$

but $h(4)$ is not defined.
As x approaches 5, the function grows without bound, so

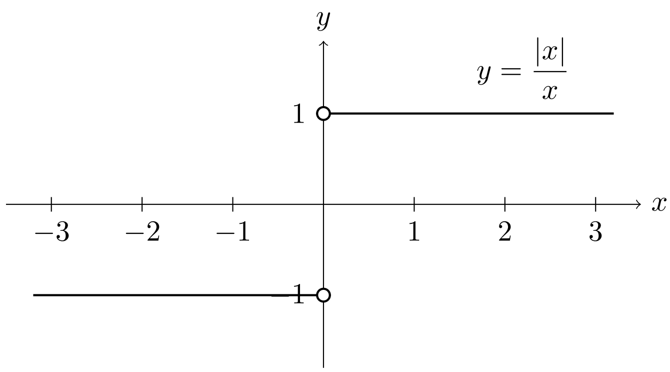
$$\lim_{x\to 5} h(x)$$

does not exist as a finite number.

1.2 Find $\lim_{x \to 0} \frac{|x|}{x}$

Answer

This is better visually but we can see that for $x > 0$ the functions is 1 and for $x < 0$ the function is -1, and at $x = 0$ the denominator is 0 and the function is undefined at that point. Therefore the limit does not exist at 0.



1.3 Find $\lim_{x \to \infty} \frac{1}{x}$

Answer

This will be simply 0, since denominator tends to 0 and the resultant fraction will tend to ∞.

1.4 Find $\lim_{x \to \infty} \frac{7x - 2}{11x + 9}$

Answer

$$\lim_{x \to \infty} \frac{7x - 2}{11x + 9} = \lim_{x \to \infty} \frac{7 - 2/x}{11 + 9/x} = \frac{7}{11}$$

1.5 Find $\lim_{x \to -\infty} \frac{4x^2 - 5x + 3}{x^2 + 2}$

Answer

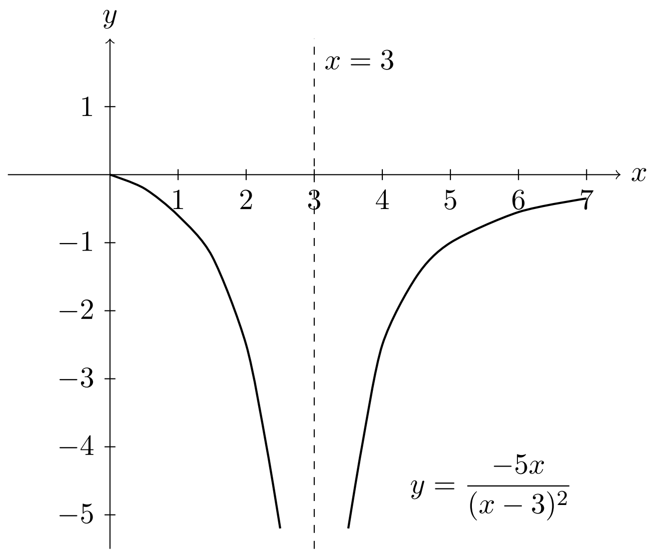
$$\lim_{x \to -\infty} \frac{4x^2 - 5x + 3}{x^2 + 2} = \lim_{x \to -\infty} \frac{4 - 5/x + 3/x^2}{1 + 2/x^2} = 4$$

1.6 Find $\lim_{x \to 3} \frac{-5x}{x^2 - 6x + 9}$

Answer

$$\lim_{x \to 3} \frac{-5x}{x^2 - 6x + 9} = \lim_{x \to 3} \frac{-5x}{(x-3)^2} = -\infty$$

As x reaches 3, the numerator tends to -15 and the denominator is very small. So at $x = -15$ we get a very large value which is negative ∞. Something like below. Ideally I should plot these in Python but I am using LLMs to generate tikz figures quickly. So please verify real plots when we have to truly plot graphs.

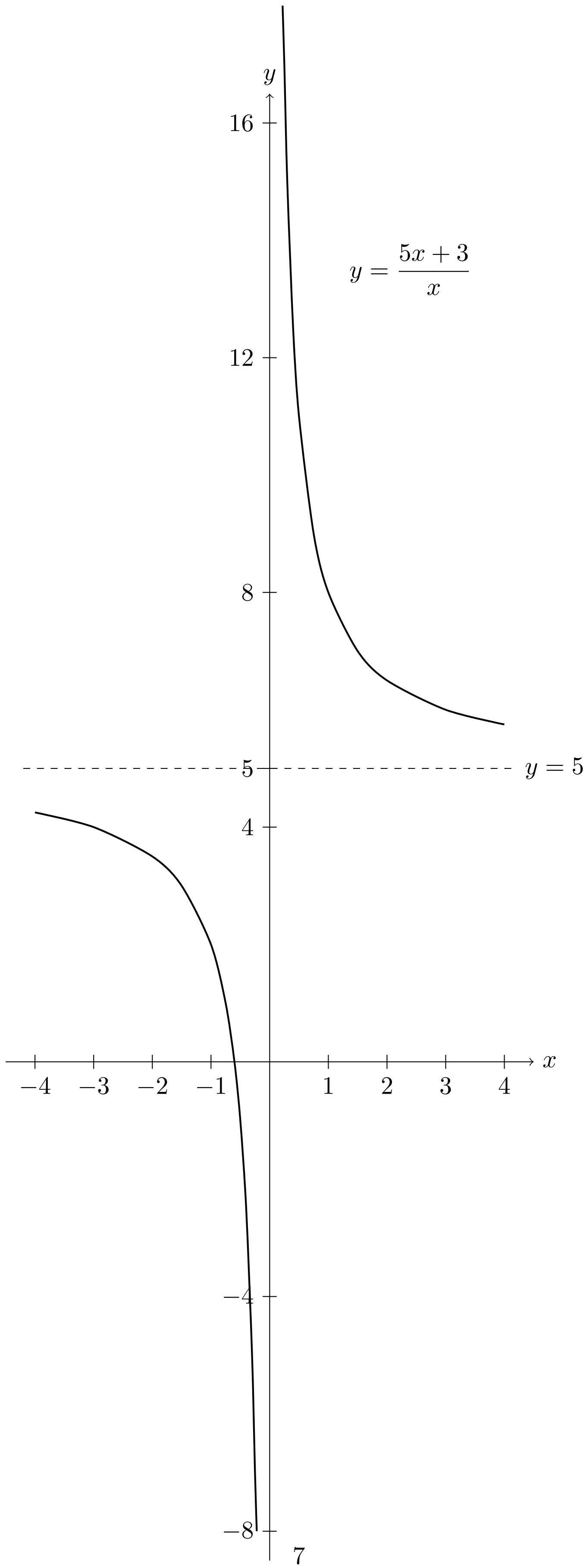


1.7 Find $\lim_{x \to 0} \frac{5x+3}{x}$

Answer

$$\lim_{x \to 0} \frac{5x+3}{x} = \lim_{x \to 0} 5 + \frac{3}{x}$$

As $x \to 0$ from the positive side of x the value of $\frac{3}{x}$ becomes very large towards $+\infty$. Similarly from the negative side of x the value becomes $-\infty$. Thus limit actually does not exist. If we were computing one sided limits we could get the values. The graph shows the curve.



y
16
12
8
5
4
-4
-8
$y = \frac{5x+3}{x}$
$y = 5$
x
-4 -3 -2 -1 1 2 3 4

1.8 Compute $(1.05)^2$

Answer

1.9 Compute $\sqrt{256.03}$

Answer

1.10 Compute $\sqrt{257}$

Answer

1.11 Compute $\sqrt{255.72}$

Answer

1.12 Compute $\sqrt{255}$

Answer

1.13 Compute $\dfrac{1}{\sqrt{0.99} + 0.99^2 + 0.99}$

Answer

19 References

Marsden, Jerrold E. and Alan Weinstein, *Calculus I, II, III.*
https://www.cds.caltech.edu/~marsden/volume/Calculus/